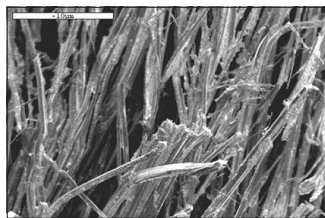
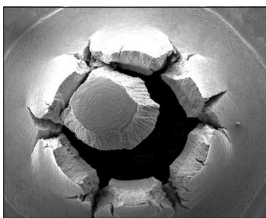


Handbook on Lead-bismuth Eutectic Alloy and Lead Properties, Materials Compatibility, Thermal- hydraulics and Technologies

2015 Edition



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and Lead Properties, Materials Compatibility,
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Foreword

Prompted by several international projects dedicated to heavy liquid metal (HLM) nuclear systems designed for the transmutation of highly active nuclear waste, the Expert Group on Heavy Liquid Metal Technologies began its activities in 2003. The accelerator-driven system (ADS) was the reference system considered for this purpose, and several ADS concepts have been developed, for example, in Europe, Japan, Korea and the United States. The general layout of an ADS foresees three main components: i) the subcritical blanket loaded with minor actinide-bearing fuel and cooled with liquid lead-bismuth eutectic (LBE) or liquid lead (Pb); ii) an external neutron source made from a liquid LBE as a spallation target; and iii) a powerful proton accelerator. Several experimental campaigns on these key components were identified in a roadmap on the demonstration of ADS feasibility, with design, construction, operation and post-operation investigations of a neutron spallation target made of liquid LBE ensuing as an important task towards the demonstration of the ADS concept. The Megawatt pilot target experiment (MEGAPIE) was launched in this context to demonstrate the feasibility of a liquid, lead-bismuth (Pb-Bi) target for spallation facilities at a beam power level of 1 MW. MEGAPIE was supported by an international consortium composed of partners from Europe, Japan, Korea, the United States and the European Commission, demonstrating the transnational interest in this technology.

In addition to ADS development, other international projects continue to advance along other paths, including within the framework of the Generation IV International Forum (GIF), which has selected the lead-cooled fast reactor (LFR) as one of its six reference concepts; and through Russian projects to develop both the lead-cooled BREST and the LBE-cooled SVBR (Svintsovo-Vismutovyi Bystryi Reactor) concepts.

These activities and initiatives motivated the Expert Group on HLM Technologies to focus on the establishment of a common database in the form of a handbook so as to deliver a reference work for engineering design purposes. While the expert group was finalising the first version of the handbook, activities on HLM were expanding worldwide. For instance, in 2005, the European Commission supported project, EUROTRANS, was initiated. It gathered key European partners with significant ADS programmes and facilitated a collaboration framework with R&D programmes supported by the United States Department of Energy (US DOE) and at the time, the Japan Atomic Energy Agency Research Institute (now Japan Atomic Energy Agency). In parallel, international R&D projects on HLM technologies in support of the GIF LFR concept were also undertaken. The expert group thus perceived the necessity to edit, in a relatively short time span, a first version of the handbook in order to gather a set of relevant data for both ADS and LFR engineering design activities, but also to identify R&D gaps and needs as input towards future R&D programmes. At the time of the first edition's publication, it was evident that these newly initiated projects were producing highly relevant data worth including in a second edition of the handbook.

The second edition of the handbook was compiled and edited over a period of approximately seven years, and it includes new data that has helped in better understanding phenomena such as liquid metal embrittlement or turbulent heat transfer in a fuel bundle. In addition, new experimental data and updated and/or alternative explanations of phenomena such as post-irradiation examination of irradiated samples

2.8 Saturation vapour pressure

The vapour pressure of a liquid metal as a function of temperature is an important property of coolants, which determines the dependence of the boiling temperature on pressure and heat-mass transfer in the cover gas space. It is directly related to the latent heat (enthalpy) of vaporisation. At the equilibrium between liquid and vapour phases on the liquid-vapour phase transition line, the Clausius-Clapeyron equation states (Iida, 1988):

$$\left(\frac{dp}{dT}\right)_{l \rightarrow v} = \frac{H_v - H_l}{T_B(V_v - V_l)} \quad (2.13)$$

where H_l and H_v are molar enthalpies, and V_l and V_v are molar volumes of the liquid and gas phases respectively. Assuming that the vapour behaves as a perfect gas and neglecting the volume of liquid in comparison with that of the gas, it can be obtained from Eq. (2.13) that the saturation vapour pressure (p_s) is:

$$p_s(T_B) = A \cdot \exp\left(-\frac{\Delta H_B}{R \cdot T_B}\right) \quad (2.14)$$

where A is a constant of integration and $\Delta H_B = H_v - H_l$ is the heat (enthalpy) of evaporation.

The above equation can provide approximate values for equilibrium vapour pressures over a wide range of temperature due to the relatively small variation of ΔH_B with temperature at low pressures. Closer fits of the experimental results can sometimes be obtained by adding supplementary temperature dependent terms, for instance, the following equation is frequently used:

$$\ln(p_s(T_B)) = A_1 + \frac{B}{T_B} + C \cdot \ln(T_B) \quad (2.15)$$

2.8.1 Lead

The data for the saturation vapour pressure of lead at different temperatures were found in the following publications: Lyon (1954, 1960), Friedland (1966), Hultgren (1973b), Iida (1988), Kubaschewski (1979, 1993), Cheynet (1996), Imbeni (1998) and in SMRB (Smithells, 1983, 2004).

Miller (1954) in Lyon (1954) used data on the Pb vapour pressure from Ditchburn (1941). Friedland (1966) presented plots based on the data from Lyon (1954). Hultgren (1973b) analysed different sources and presented a table with the recommended data in the range from the melting to the boiling temperature. Kubaschewski (1979) collected the data from Gross (1948), Eucken (1936) and Kelley (1935) and presented their own correlation. Iida (1988) reproduced the correlation of Kubaschewski (1979). Cheynet (1996) recompiled results communicated by himself earlier (1989) and by Chase (1985), Barin (1989) and Knacke (1991). Imbeni (1998) and SMRB (Smithells, 1983, 2004) recommended using the correlation of Kubaschewski (1979). The selected data sources are described in Table 2.8.1.

Figure 2.8.1(a) shows the saturated vapour pressure of molten lead taken from the above-cited literature and plotted as a function of temperature. At first glance, it can be seen that the agreement among the different sources is satisfactory. The correlation proposed by Kubaschewski and Alcock renormalised to obtain $p_s = 0.1013$ MPa at $T_{B,0(\text{Pb})} = 2021$ K is thus recommended in Sobolev (2011) for the saturated vapour pressure of liquid lead. It can be presented as follows (where temperature is in kelvins):

$$p_{s(\text{Pb})}[\text{Pa}] = 1.88 \times 10^{13} \cdot T^{-0.985} \cdot \exp(-23325/T) \quad (2.16)$$

The deviation of the selected literature data on the saturated vapour pressure of liquid lead from the recommended correlation (2.16) is plotted in Figure 2.8.1(b) (reproduced from Sobolev [2011]). One can see that, except for old data from Miller (1954), agreement is rather good: the deviation at temperatures higher than 1 100 K is less than 5% and increases to about 15% close to the melting temperature.

Using the selected data and Eq. (2.14) the following simplified correlation was also obtained (Sobolev, 2011):

$$p_{s(\text{Pb})}[\text{Pa}] = 5.76 \times 10^9 \cdot \exp(-22131/T) \quad (2.16a)$$

The deviation of the selected database from the correlation (2.16a) is up to 18%.

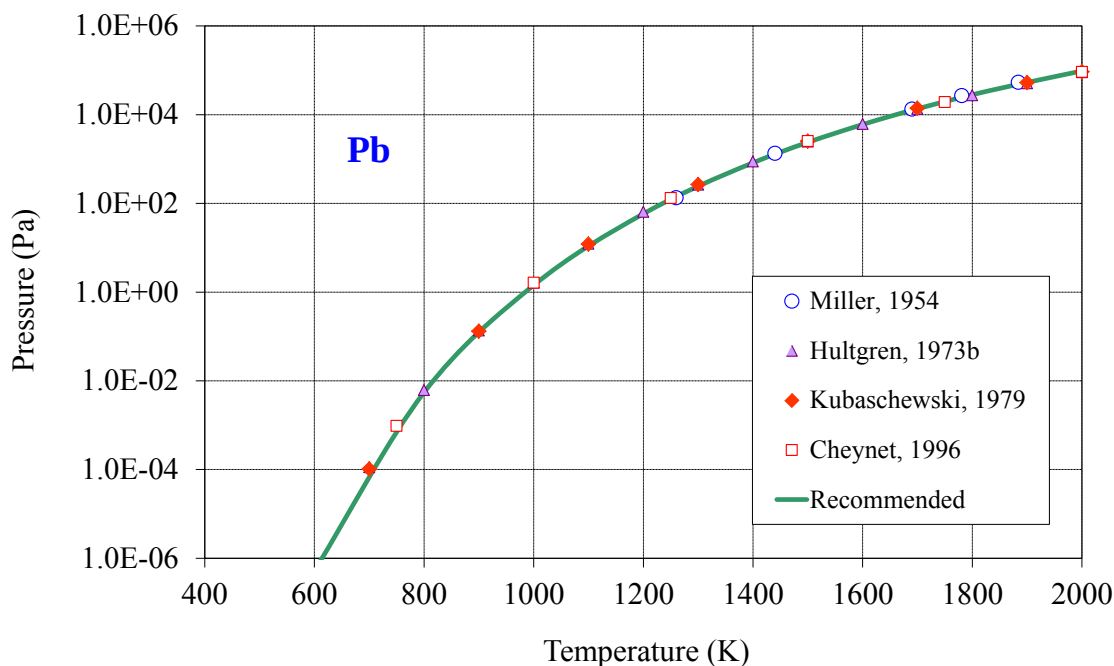
2.8.2 Bismuth

The data on the saturation vapour pressure of liquid bismuth as a function of temperature were found in the following handbooks and compilations: Lyon (1954, 1960), Hove (1961), Hultgren (1973b), Kubaschewski (1979, 1993), Cheynet (1996), Imbeni (1998a) and SMRB (Smithells, 1983, 2004).

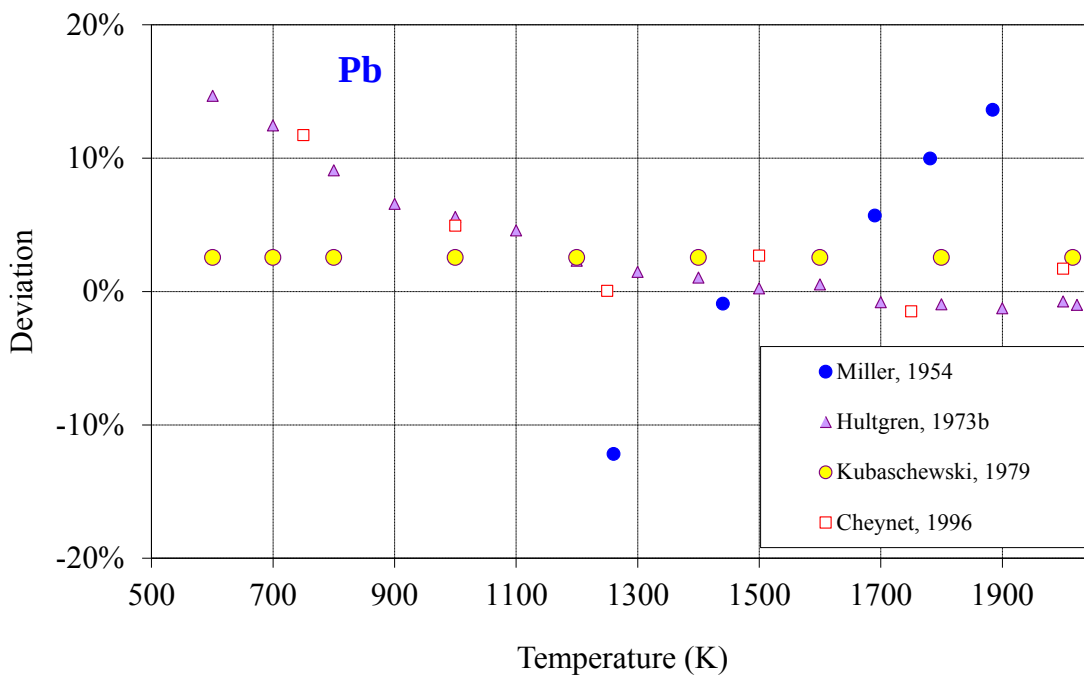
Miller (1954) in Lyon (1954) used data on the Bi vapour pressure from Ditchburn (1941). Hove (1961) based his compilation on the data presented in Lyon (1954), Smithells (1955), Hansen (1958) and other older sources. Hultgren (1973b) reported values selected on the basis of a critical assessment of many previously published data. Kubaschewski (1979) presented correlations for vapour pressure of Bi and Bi₂ with reference to a report of Brackett (1957). Iida (1988) reproduced the correlations of Kubaschewski (1979). Cheynet (1996) recompiled results communicated by himself earlier (1989) and by Chase (1985), Barin (1989) and Knacke (1991). Imbeni (1998) and SMRB (Smithells, 1983, 2004) recommended using the correlations of Kubaschewski (1979). The selected data are described in Table 2.8.2.

Table 2.8.1: Selected recommendations on the saturated vapour pressure of liquid lead

Reg. No.	Parameter	Measurement method	Estimated accuracy	Temperature range, K	Pressure range, Pa	Values	Ref.
8.1.1	Saturated pressure	?	?	1 260-1 884	10 ² -10 ⁵	–	Miller (1954)
	Interpolation function: $p_s = 5\,863 \times 10^9 \cdot \exp(-22\,240/T)$						
	Comments: Unknown purity						
8.1.2	Saturated pressure	n/a	?	601-2 100	10 ⁻⁷ -10 ⁵	–	Hultgren (1973b)
	Interpolation function: $p_s = 7.994 \times 10^9 \cdot \exp(-22\,338/T)$						
	Comments: Unknown purity						
8.1.3	Saturated pressure	n/a	?	601-2 023	10 ⁻⁷ -10 ⁵	–	Kubaschewski (1979)
	Interpolation function: $\log(p_s) = 2.1249 - 10\,130/T - 0.985 \cdot \log(T) + 11.16$						
	Comments: Unknown purity						
8.1.4	Saturated pressure	n/a	?	750-3 000	10 ⁻³ -10 ⁶	–	Cheyne (1996)
	Interpolation function: $p_s = 5.343 \times 10^9 \cdot \exp(-21\,955/T)$						
	Comments: Unknown purity						

Figure 2.8.1(a): Saturated vapour pressure of liquid lead versus temperature

Source: Sobolev (2011).

Figure 2.8.1(b): Deviation of the selected literature data on the saturated vapour pressure of liquid lead from the recommended correlation (2.16)

Source: Sobolev (2011).

Table 2.8.2: Selected recommendations on the saturated vapour pressure of liquid bismuth

Reg. No.	Parameter	Measurement method	Estimated accuracy	Temperature range, K	Pressure range, Pa	Values	Ref.
8.2.1	Saturated pressure	?	?	1 190-1 673	10 ² -10 ⁴	–	Miller (1954)
	Interpolation function: $p_s = 1.388 \times 10^{11} \cdot \exp(-24\,722/T)$						
	Comments: Unknown purity						
8.2.2	Saturated pressure	?	?	1 193-1 773	10 ² -10 ⁴	–	Hove (1961)
	Interpolation function: $p_s = 1.137 \times 10^{10} \cdot \exp(-21\,925/T)$						
	Comments: Unknown purity						
8.2.3	Saturated pressure	n/a	?	545-2 000	10 ⁻⁸ -10 ⁵	–	Hultgren (1973b)
	Interpolation function: $p_s = 2.435 \times 10^{10} \cdot \exp(-22\,621/T)$						
	Comments: Unknown purity						
8.2.4	Saturated pressure	n/a	?	545-1 833	10 ⁻⁸ -10 ⁵	–	Kubaschewski (1979)
	Interpolation function: $\log(p_{s(\text{Bi1})}) = 2.1249 - 10\,400/T - 1.26 \cdot \log(T) + 12.35$ $\log(p_{s(\text{Bi2})}) = 2.1249 - 10\,730/T - 3.2 \cdot \log(T) + 18.1$						
	Comments: Unknown purity						
8.2.5	Saturated pressure	n/a	?	750-2 000	10 ⁻³ -10 ⁵	–	Cheyne (1996)
	Interpolation function: $p_s = 2.016 \times 10^{10} \cdot \exp(-22\,363/T)$						
	Comments: Unknown purity						

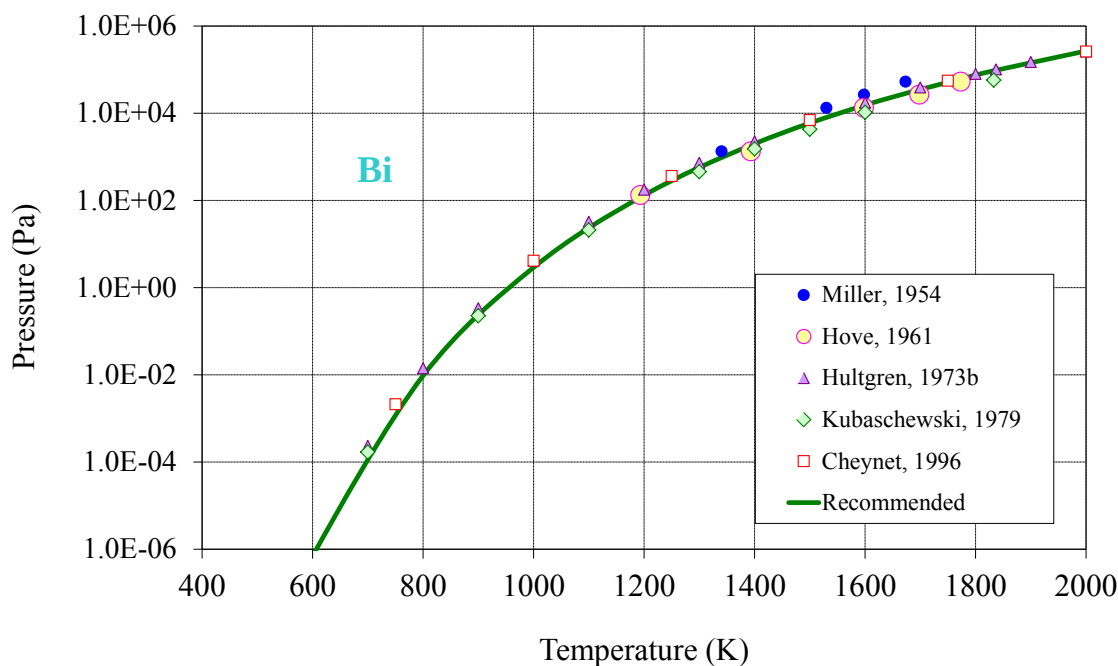
Figure 2.8.2(a) shows the values for the saturated vapour pressure of liquid bismuth from the above-cited sources plotted as a function temperature.

It can be seen that the agreement in temperature dependence given by the different sources is satisfactory. However, a significant scatter exists in values. Therefore, Eq. (2.14) was chosen to construct an empirical correlation allowing calculation of the saturated vapour pressure of liquid bismuth as a function of temperature. Using the selected data, the following correlation was obtained in Sobolev (2011), where temperature is in kelvins:

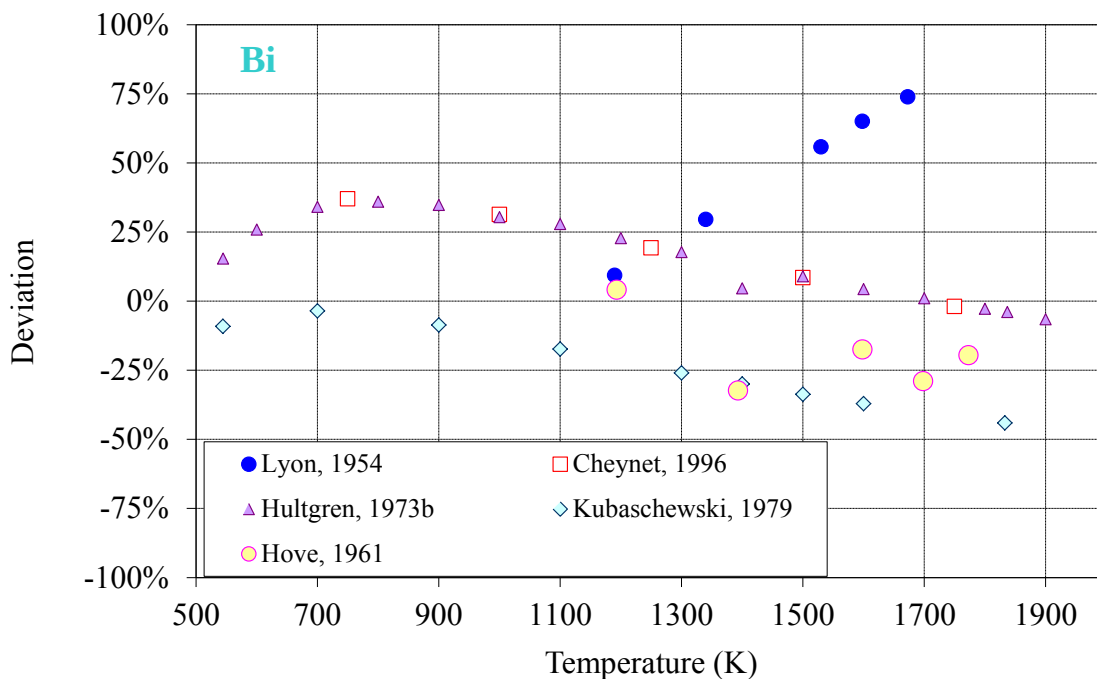
$$p_{s(\text{Bi})} [\text{Pa}] = 2.67 \times 10^{10} \cdot \exp(-22858/T) \quad (2.17)$$

This correlation is recommended for temperatures up to the normal boiling point.

The deviation of the selected literature data for the saturated vapour pressure of liquid bismuth from the recommended correlation (2.17) is illustrated in Figure 2.8.2(b) (reproduced from Sobolev [2011]). One can see that a very large difference exists between the selected sources. The recommended correlation (2.17) gives the “best estimate” values with a maximum uncertainty of about 40%, if the earliest data presented in Miller (1954) are disregarded. Under these circumstances, it is not possible to develop a more precise correlation.

Figure 2.8.2(a): Saturated vapour pressure of liquid bismuth versus temperature

Source: Sobolev (2011).

Figure 2.8.2(b): Deviation of the selected literature data on the saturated vapour pressure of liquid bismuth from the recommended correlation (2.17)

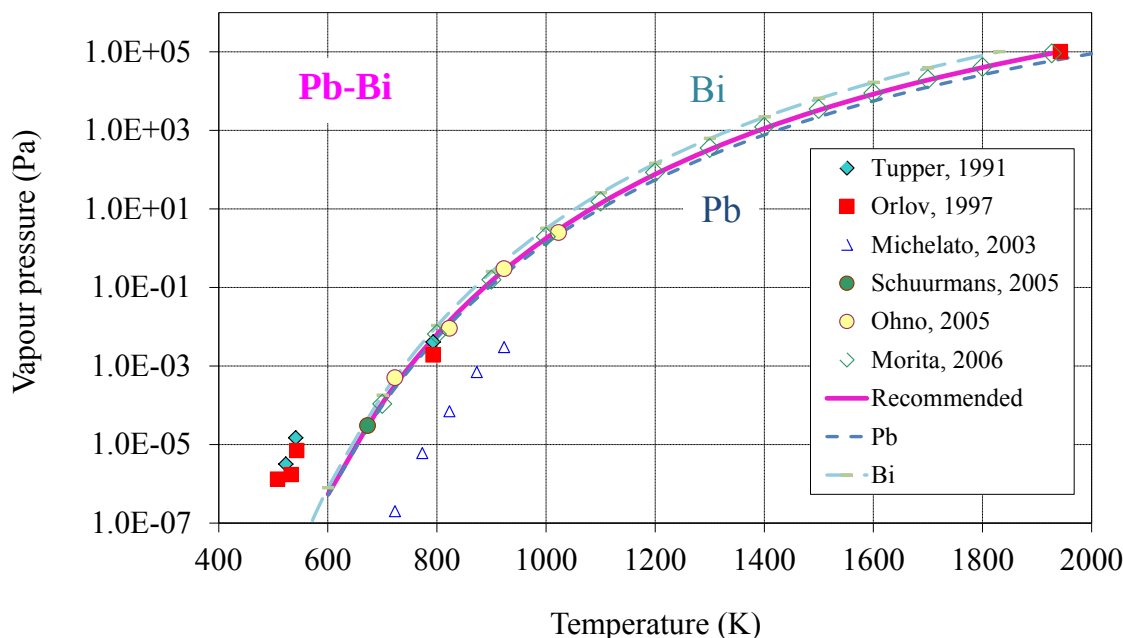
Source: Sobolev (2011).

2.8.3 LBE

Tupper (1991) studied polonium evaporation from LBE and published some experimental results on the LBE saturated vapour pressure at temperatures 235-268 and 520-550°C (508-541, 793-823 K). Later, Orlov (1997) published experimental values for the saturation pressure of the LBE vapour at equilibrium with the liquid phase at five different temperatures. However, both publications mentioned that a very large uncertainty in the pressure measurement exists at low temperatures ($< 500^{\circ}\text{C} \sim 773\text{ K}$). The results of Orlov (1997) were reproduced later in Gromov (1999) and Cevolani (1998). New results have been communicated by Michelato (2003), Schuurmans (2005), and published by Ohno (2005). Cevolani (1998) fit a four-term logarithmic formula to the experimental data of Orlov (1997). Imbeni (1999) also used data of Orlov (1997), but recommended more simple correlations of type (2.14) and (2.15). Morita (2006) proposed a four-term correlation, which takes into account the formation of Bi_2 molecules in the LBE vapour phase. The sources included in the database are presented in Table 2.8.3 and their results are plotted in Figure 2.8.3(a).

Table 2.8.3: Selected database on the saturated vapour pressure of liquid LBE

Reg. No.	Parameter	Measurement method	Estimated accuracy	Temperature range, K	Pressure range, Pa	Values, Pa	Ref.
8.3.1	Saturated pressure	Evaporation/condensation	?	523-823	10^{-6} - 10^{-2}	–	Tupper (1991)
	Interpolation function: $p_s = 4.816 \times 10^7 \cdot \exp(-16\,429/T)$						
	Comments: 55.5 wt.% Bi + 44.5 wt.% Pb						
8.3.2	Saturated pressure	?	?	508-793	10^{-6} - 10^{-3}	–	Orlov (1997)
	Interpolation function: $p_s = 1.207 \times 10^8 \cdot \exp(-16\,956/T)$						
	Comments: 55.5 wt.% Bi + 44.5 wt.% Pb						
8.3.3	Saturated pressure	(Direct)	?	723-923	10^{-7} - 10^{-3}	–	Michelato (2003)
	Interpolation function: $p_s = 7.97 \times 10^{11} \cdot \exp(-30\,597/T)$						
	Comments:						
8.3.4	Saturated pressure	Direct	(10%)	673.15	n/a	3×10^{-5}	Schuurmans (2005)
	Interpolation function: $p_s = 1.138 \times 10^{10} \cdot \exp(-22\,597/T)$						
	Comments:						
8.3.5	Saturated pressure	Transpiration	5-25%	723-1 023	10^{-4} - 10^0	–	Ohno (2005)
	Interpolation function: $p_s = 3.015 \times 10^9 \cdot \exp(-21\,447/T)$						
	Comments:						
8.3.6	Saturated pressure	n/a	(30-60%)	700-1 943	10^{-4} - 10^5	–	Morita (2006)
	Interpolation function: $\ln(p_s) = 35.773 + 2.8006 \times 10^{-4} \cdot T - 24\,053/T - 1.6402 \cdot \ln(T)$						
	Comments: 55.5 wt.% Bi + 44.5 wt.% Pb						

Figure 2.8.3(a): Saturated vapour pressures of liquid LBE versus temperature

Source: Sobolev (2011).

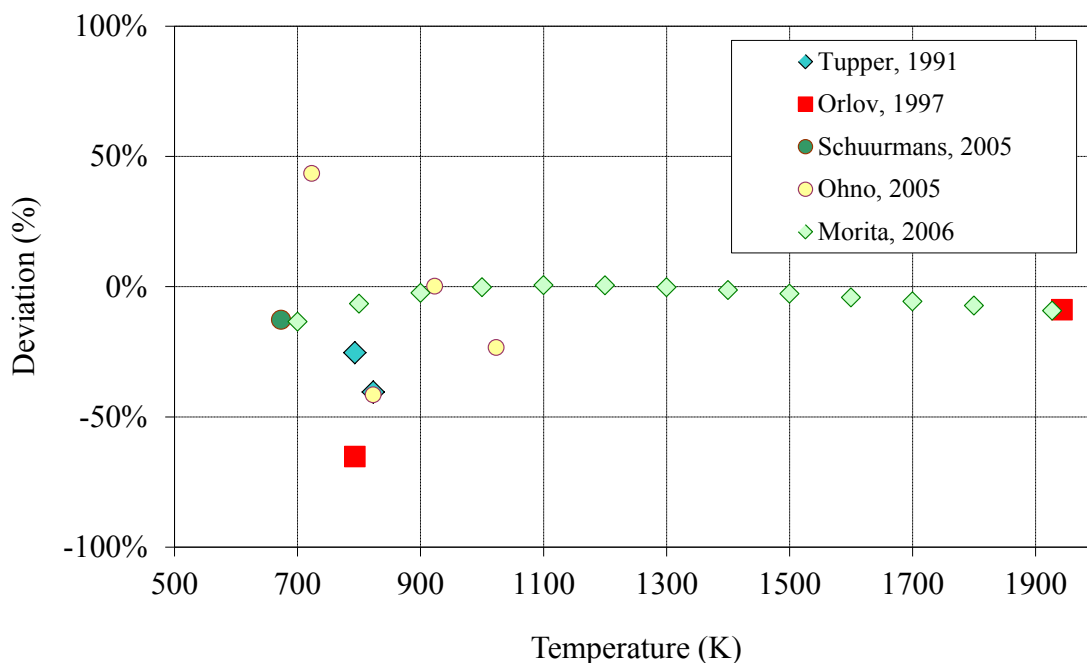
Figure 2.8.3(a) illustrates that in the temperature range 650-1940 K, the experimental values for LBE vapour pressure available from Tupper (1991), Orlov (1997), Schuurmans (2005) and Ohno (2005) are very close. The results of Michelato (2003) are systematically lower than the body of data. At temperatures of 500-550 K (227-277°C), Tupper (1991) and Orlov (1997), give a LBE vapour pressure that is an order of magnitude higher than the liquid-vapour equilibrium lines of lead and bismuth extrapolated to the lower temperatures. It should be noticed that at the expected level of saturated pressure ($\sim 10^{-8}$ Pa) it is very difficult to measure the pressure correctly, and overestimation by several orders of magnitude can often be obtained due to contamination of the liquid metal or of the experimental rig. Therefore it is proposed to not consider these points at present.

The selected data set was used to determine the coefficients for a simplified equilibrium Eq. (2.14), taking into account $p_s \approx 0.1$ MPa at the normal boiling temperature. The values of ΔH_B and A thus obtained were: $\Delta H_B = 187.5$ kJ mol $^{-1}$ and $A = 12.2 \times 10^9$ Pa (Sobolev, 2011). The obtained value of ΔH_B is in a good agreement with the value of the latent heat of LBE vaporisation at the normal boiling point recommended in Section 2.7.3 (178 kJ mol $^{-1}$), taking into account the large uncertainty in the experimental results used for its calculation. The following simplified correlation for the saturation vapour pressure of molten LBE is recommended, where temperature is in kelvins:

$$p_{s(\text{LBE})}[\text{Pa}] = 1.22 \times 10^{10} \cdot \exp(-22552/T) \text{K} \quad (2.18)$$

This correlation is similar to those proposed above for lead and bismuth. The deviation of the selected experimental values from this correlation is illustrated in Figure 2.8.3(b). The maximum deviation in the temperature range of 673-1927 K (400-1654°C) is about $\pm 60\%$. (There is no agreement at all at the lower temperatures.) In spite of this significant deviation, it was decided not to search for a more sophisticated formula, given the very large dispersion of data obtained from different sources.

Figure 2.8.3(b): Deviation of the selected data on the saturated vapour pressures of liquid LBE from the recommended correlation (2.18)



Source: Sobolev (2011).

2.9 Critical point parameters

The critical temperature (T_c), the critical pressure (p_c) and the critical density (ρ_c) (or the critical specific or molar volume V_c) are very important parameters for the development of an equation of state (EOS) and the extension of information on the available thermophysical data to higher temperatures and pressures. A very large uncertainty still exists in the critical temperatures, pressures and densities of heavy metals of interest. The critical compressibility Z_c is often used to compare different substances. It is determined as follows:

$$Z_c = \frac{p_c \cdot V_c}{R \cdot T_c} = \frac{p_c \cdot M}{R \cdot T_c \cdot \rho_c} \quad (2.19)$$

where M is the molecular mass.

The following sources were analysed to prepare recommendations for the critical parameters of Pb, Bi and LBE in this handbook: Grosse (1961), Cahill (1968), Fortov (1975), Pottlacher (1990, 1998), Martynyuk (1983, 1998), Azad (2005), Morita (2006, 2007), Kirillov (2008) and Sobolev (2008a, 2008b, 2009, 2011).

2.9.1 Lead

Many experimental results were obtained for the Pb critical parameters, but with a large dispersion.

Grosse (1961) estimated the Pb critical temperature $T_{c(\text{Pb})} \approx 5\,400\text{ K}$ ($2\,127^\circ\text{C}$) with the principle of corresponding states using mercury (Hg) as the reference fluid and the enthalpy of the Pb vaporisation $\Delta H_{B,0}$ at the normal boiling point from an earlier source (Stull, 1956).